# Vertically Integrated Industries

Investment theses and category views on minerals, metals, manufacturing and chemistry

## 1. Investment theses: tailwinds, key takeaways and macro

This theme covers companies that make physical things in their own plants, such as metal parts, magnets and molecules, and that use their own software and AI to make them faster and with fewer skilled people than older competitors. They are vertically integrated because one company owns the whole job, from order to shipped product, and earns money on what it ships. The five theses below explain why demand is rising, where new companies have an edge, how buyers behave, what the unit economics look like, and where the main risk sits.

1. **The US is moving its supply of minerals, magnets and parts away from China for national security and economic resilience, and defense budgets, government lending and a lasting labor shortage will keep that demand in place for years**
   - China leads refining in 19 of 20 strategic minerals, with about 70% average share (IEA), and makes about 94% of the world's rare-earth magnets, which go into drones, missiles, motors and robots (Mining.com). The US relies fully on imports for 16 critical minerals (USGS), and China has already used export controls on these materials; its current pause on those controls ends on November 10 and November 27, 2026. Buyers and the government will pay for non-Chinese supply even when it costs more, because supply chain resilience is now an economic and national-security issue.
   - Defense budgets are rising and the Pentagon is paying to rebuild domestic supply of munitions and parts, with the FY27 request asking for $49.5B for 10 key munitions, about 8.5 times the average of the past decade (CNAS), and the Navy putting $900M into submarine-parts plants (Breaking Defense). Because the FY26 defense bill also lets the Pentagon commit to purchases of up to 10 years with advance payments (FY26 NDAA), factories that qualify for defense work can count on years of steady orders.
   - The government now lends to and invests in the companies rebuilding these supply chains, which lowers the cost of building new plants. The Pentagon's Office of Strategic Capital (OSC), which lends to companies in key supply chains, has made a $620M loan to Vulcan Elements, a US magnet maker, and a $500M conditional loan to Phoenix Tailings, a US rare-earth metals maker, and about $20B is requested for it in FY27 (Mintz). The Department of Energy's loan office has over $289B of lending authority (DOE), and governments worldwide put about $65B into minerals in 2025 (IEA).
   - US manufacturers will stay short of skilled workers for years, since they need to fill about 3.8M jobs by 2033 and about 1.9M of those jobs are projected to go unfilled (Deloitte), which favors plants that can run with fewer people. Of the 17,156 machine shops in the US, 83% have fewer than 20 employees (Census), and many owners are near retirement with no successor.
   - Some of the current tailwinds are temporary, so we should not build the thesis on them. Tariffs are set by proclamation and can change overnight, about 82% of the munitions request depends on a reconciliation bill Congress has not passed (CNAS), and government loans take a long time to close and come with conditions. The Fed raised rates to 3.75-4.00% in September, and spending on new factory construction was down 21.9% year on year by May 2026 (Federal Reserve, ENR). We should favor companies that still work if tariffs fall and the munitions surge slows.
2. **AI and automation let new companies quote, program and inspect parts faster and get more output from plants that run nonstop, which gives them an edge over older competitors**
   - The work around the machine includes quoting, planning, writing the instructions a machine follows (CAM programming), inspection and paperwork. Arda, a factory software startup, says its AI cut the time to write assembly instructions at Anduril from about 10 hours to 15 minutes (claim), and a former Hadrian employee told us in an expert call that Hadrian cut its lead times by about 50% through automation.
   - The machines in most plants sit idle for much of the day, which leaves room to grow output without building anything new. In a survey of more than 100 plants, cutting machines ran about 34% of the time

against a 70% target, and European process plants run at about 74% of capacity against the 80-85% they need (FTI Consulting), a gap that planning and control software can close without a new plant.

- In a plant that runs nonstop, such as a chemical or metals line, every extra point of yield (good output per unit of input) is close to pure profit. Labor is under 10% of the cost of goods in a new automated plant (claim), so the case for automation rests on speed and yield more than on headcount.
- Nobody has yet shown that AI makes each unit cheaper as volume grows, so the edge today is speed, yield and the ability to grow with a smaller crew while competitors cannot hire.

3. **Defense and aerospace buyers pay a premium for fast delivery and spare capacity, and a supplier that wins approval keeps the work for decades**
   - Defense buyers care more about delivery dates than price, and a director at a defense prime told us in an expert call that "cost is secondary" because it is "driven usually by schedule", while an engineer there said he would pay 20-40% more to avoid a delay. The premium is highest on prototypes, spare parts and urgent orders, and it fades at volume.
   - A major defense contractor, often called a prime, approves a supplier only after the plant exists and is certified, which takes about nine months to two years (expert calls with two primes). Once approved, a supplier keeps the work for aerospace programs that can run about 50 years.
   - The Pentagon relies on more than 200,000 suppliers and has country-of-origin data for fewer than 10% of the lower-tier suppliers on the F-35 (GAO). Most of that supplier base is small, old and short of people, so primes need new suppliers that can deliver on time.
4. **Companies that own their plants earn the best margins, and the strongest ones get governments, customers or lenders to fund much of the build**
   - Companies that own their plants keep more of each sale than those that route work to partners, and Protolabs, a listed maker of custom parts, earns a 49% gross margin in its own plants and 31% on work it sends to partner shops (company filings). Approved incumbents also earn well, and Howmet Aerospace's engine-parts unit ran a 37.7% EBITDA margin (earnings before interest, taxes, depreciation and amortization) in the second quarter of 2026 (company filings).
   - The central risk is capital spending (capex) on buildings, machines and certifications before contracts are signed, because the money goes out years before revenue comes in. Hadrian, a precision-parts maker, committed about $200M to a single plant in Mesa, Arizona (company release), and its CEO says capex runs about two years ahead of revenue (claim).
   - The companies that succeed shift much of the cost of the build to others, using government loans and grants, purchase agreements signed before the plant is built, factories hosted inside customers' plants, bank debt once revenue exists, leased buildings, franchising, and acquisitions of certified plants.
5. **Established incumbents control supplier approvals and buy promising startups, but an aging and fragmented supplier base still leaves room for new companies**
   - The primes decide which suppliers are approved and how much work each one gets, and they cap each supplier at about a quarter to a third of their work, so even a winning supplier grows one share at a time. Precision Castparts, one of the largest aerospace parts suppliers, saw its share of Boeing's machining work fall from about 50% to 30% over ten years, and it is now automating (expert call).
   - Incumbents often buy new technology instead of building it, which gives startups buyers but also well-funded competitors. BHP's venture arm backs Mariana Minerals, Rio Tinto paid $6.7B for Arcadium Lithium, and RTX Ventures, the venture arm of the defense contractor RTX, led the seed round of Radical AI, an AI materials lab. Drug companies are also building their own AI labs, and Eli Lilly and NVIDIA plan to invest up to $1B over five years in a joint AI lab for drug discovery in the San Francisco Bay Area (Lilly and NVIDIA, January 2026).
   - Hardware companies in this space have failed when their growth ran ahead of their economics, as Markforged, a 3D-printer maker, fell from a $2.1B valuation in 2021 to a $42.5M sale in 2026 (3D Printing Journal), and Daedalus, an AI-run machining startup in Germany, went insolvent in August 2026.

- New companies win where the incumbents are weakest, among small, aging shops with no successor, new defense programs that need second sources, and plants that need software to run with fewer people.

## 2. Interesting categories

The categories below follow our primer, and for each one we set out the main tailwinds and headwinds and state whether we should focus there, judged on the category as a whole.

### A. Raw Materials

#### Stage A: Locating & finding minerals and mines

- **Tailwinds: Cutting reliance on China starts with finding new deposits of copper, lithium, nickel and rare earths in the US and allied countries.** The US relies fully on imports for 16 mineral commodities and for more than half of 54 others, including 67% of its rare earths and 57% of its refined copper (USGS).
- **Demand for these metals is rising with batteries, magnets, the grid and data centers.** Copper demand alone is expected to rise about 50% by 2040 as these uses grow (Robeco).
- **AI combined with cheaper drilling and faster lab testing lowers the cost of checking each exploration target.** The major miners are paying for these new methods through their venture arms, which shows that buyers for the work exist.
- **Headwinds: Most exploration projects never become producing mines, and the few that do take 15 to 19 years.** Only about 0.5% of exploration projects succeed by one estimate (claim), and mines average 15.7 years from discovery to production, US mines opened in 2020-24 took 19.1 years, and US permitting alone commonly takes 7-10 years, against about 2 in Australia and Canada.
- **Each discovery is expensive for the industry as a whole, and few of the deposits found become mines.** The industry spends about $300M on exploration per deposit found, and only about a fifth of the roughly 40 deposits found each year become mines (Earth AI, claim).
- **Why we should not focus here: Explorers rarely keep the value of what they find, because the miner that builds the mine takes most of the economics.** Explorers usually bring in a miner through an option or joint venture, in which the miner pays staged cash and shares and funds the drilling to earn 51-70% of the project. The explorer keeps a minority stake or a royalty, usually a net smelter return (NSR) of 1-3% of metal sales after transport (industry option terms). An explorer that keeps the mine must raise billions, since a new underground copper mine can cost over $2B (Ecofin), and then wait 15 to 19 years for production. In both cases the return is too small, arrives too late or needs too much capital to justify an early equity investment. Earth AI says 80 software pilots it ran with miners did not turn into recurring revenue (claim).

**Example companies:** Earth AI, KoBold Metals, Terra AI, GeologicAI, Ideon

#### Stage B: Mining & refining

- **Tailwinds: China controls most of the world's refining and magnet making, so buyers and governments are paying to build that capacity outside China.** China leads refining for 19 of 20 strategic energy minerals and makes over 90% of rare-earth magnets (IEA, Building the Base).
- **China has already used export controls on these materials, which showed buyers the price of depending on it.** Germanium reached a record ¥28,500 per kilogram in China on October 26, 2026, up about 110% for the year (East Asia Brief). Planned non-Chinese magnet capacity covers only about a third of expected 2035 demand (IEA), so buyers will keep paying for new supply.
- **Government money lowers the cost of building new refineries and magnet plants in the US.** Pentagon loans and 10-year purchase commitments with price floors support new plants, and the 45X tax credit covers about 10% of production cost until it phases out in 2031-34 (CRS).

- **The US is rebuilding its own uranium enrichment because a 2024 law bans Russian enriched uranium and new reactors need a fuel that only Russia has sold at commercial scale.** The Prohibiting Russian Uranium Imports Act bans imports of Russian low-enriched uranium (LEU), the fuel for today's reactors, and all waivers end by January 1, 2028 (Public Law 118-62, DOE). Foreign enrichers supplied about 77% of the enrichment bought by US reactor operators in 2025, and Russia alone supplied about 26% (EIA). Most advanced reactor designs need HALEU (high-assay low-enriched uranium, enriched to between 5% and 20%), which no US supplier yet sells and which has been commercially available only from Russia (DOE, Reuters). In January 2026 the Department of Energy awarded $2.7B of contracts to rebuild domestic enrichment, including $900M to General Matter, a startup founded in 2024 that is building a plant in Paducah, Kentucky and says it will enrich uranium by the end of the decade (DOE, claim).
- **Headwinds: Western plants take longer and cost more to build than Chinese ones, since Chinese refineries take about 6 months to commission, against 2 to 4 years in the West.** A Western plant takes about 5 years to build and another 3 to 5 to ramp after permits (claim), and it costs hundreds of millions of dollars before it earns revenue.
- **China sets the price for many of these materials and can push it down whenever it wants to slow Western competitors.** MP Materials, the largest US rare-earth producer, lost $85.9M on $224.4M of revenue in 2025 (company filings), so a US producer without a price floor carries the full price risk.
- **Why we should focus here: Mining and refining have the most lasting demand in the theme, because the need for non-Chinese metal and magnets comes from national security and supply chain resilience.** New companies have a real edge here, because restarting old mines that already have permits shortens the timeline, processing scrap and mine waste avoids a new mine altogether, and AI-run process control raises yield in plants that run around the clock. The category is capital-heavy, so the companies that build here pair equity with project debt and long purchase contracts.

**Example companies:** Mariana Minerals, BHP, Rio Tinto, Caterpillar, Vulcan Elements, Phoenix Tailings, ReElement, Ucore, Energy Fuels, MP Materials, General Matter, Hertha, Cyclic, Mass Magnetics, Duranium, Boston Metal, Electra, Ceibo, Nucor, Carpenter Technology

## B. Manufacturing

### Metal Supply

- **Tailwinds: Tariffs, buy-American rules and reshoring are raising demand for US metal from machine shops, fabricators and primes.** Since April 2026 the Section 232 tariffs have charged 50% on imported metal articles and 25% on goods that contain steel, aluminum or copper (White & Case), which pushes buyers toward domestic stock.
- **Headwinds: Established distributors already serve this market well and are gaining share without AI, which leaves little room for a new entrant.** The largest is Reliance, Inc., with $14.3B of net sales in 2025, about 310 locations and about 17% of all tons sold by US service centers (Reliance 10-K). Reliance grew its tons sold by 6.2% in 2025 while the industry shrank by 1.0%.
- **The second largest distributor is consolidating to compete with Reliance on size and buying power.** Ryerson, with $4.6B of net sales in 2025, which merged with Olympic Steel in February 2026 to reach about 160 facilities (Ryerson results).
- **Distribution earns thin margins even at the largest scale, which leaves a newcomer little room to pay for technology.** Distributors earn a spread on the metal plus cutting fees and hold much of their cash in inventory, so even at Reliance's scale the pretax margin was 6.8% in 2025, and Ryerson posted a net loss.
- **Why we should not focus here: The margin in metal supply stays with the incumbents, who already have the stock, the buying power and the customers.** A new distributor ties up its capital in inventory and competes on price against bigger incumbents. The main tailwind is a tariff, and a margin that rests on a tariff can disappear with one policy change.

**Example companies:** Nox Metals, Reliance, Ryerson

### Fabrication

- **Tailwinds: Reshoring and the growth of US hardware companies are creating demand for metal parts made quickly in the US.** Fabrication means cutting, bending and welding metal into frames and enclosures, and assembling wire harnesses (the bundles of wires that carry power and signals through an aircraft or missile). Defense, drone, space and energy companies buy these parts and pay extra for speed.
- **Reshoring is bringing new assembly plants to the US, and each one needs local suppliers of frames and harnesses.** Companies announced about 247,000 US jobs from reshoring and foreign investment in 2025 (Reshoring Initiative).
- **The shortage of welders and machinists favors plants that automate cutting and bending instead of hiring more people.** Software for quoting, nesting (packing parts onto a sheet to cut scrap) and scheduling lets a small crew run many machines.
- **Headwinds: The speed premium fades at volume, and US shops beat Chinese prices on only part of their work (claim).** A former Anduril engineer told us that parts from Taiwan cost only about 20-30% less than US parts and arrive more reliably (claim).
- **Reshoring announcements run well ahead of real orders, so fabricators cannot plan capacity on headlines.** Only 14 of 64 conditional reshoring announcements had turned into actual projects by mid-2026 (Reshoring Initiative).
- **Customers that grow often buy their own machines and take the work in-house.**
- **Why we should focus here: Fabrication plants cost far less than precision plants, so companies in this category can grow on their own revenue and reach profit early.** Buyers that pay for speed, low capital needs and early profits make this one of the most attractive categories in the theme.

**Example companies:** SendCutSend, 1872, Senra Systems, Machina Labs, CircuitHub, Jeevy, Protolabs

### Precision & Defense Hardware

- **Tailwinds: The Pentagon needs a larger base of trusted domestic suppliers for precision parts, the metal or composite parts made to very tight limits by machining, 3D printing, casting or molding that go into missiles, jets, satellites and submarines.** Today the Pentagon has country-of-origin data for fewer than 10% of the lower-tier suppliers on the F-35 (GAO), and munitions and submarine budgets are rising, with purchase commitments of up to 10 years now allowed. As the owners of small certified shops retire, their capacity is coming up for sale.
- **Commercial aerospace adds a second and steadier source of demand, because Airbus and Boeing together hold orders for nearly 15,000 aircraft and parts supply sets how fast they can build.** Airbus ended 2025 with a record backlog of 8,754 aircraft and Boeing with more than 6,100 (company results). A Pratt & Whitney engine shortfall delays Airbus's main jet program from reaching 70 to 75 a month until the end of 2027 (Airbus FY2025 results). Suppliers are already growing on this demand, and Howmet's revenue rose 24% in the second quarter of 2026, with commercial aerospace up 28%, defense aerospace up 11% and gas turbines up 38% (Howmet results).
- **Power, chip and space hardware need the same precision parts and are growing as fast as aerospace.** GE Vernova has sold out its gas turbines through 2028 and had 122 GW under contract in the third quarter of 2026 (GE Vernova), chip equipment sales are forecast to reach a record $165.9B in 2026, up 23% (SEMI), and the world made a record 329 orbital launch attempts in 2025 (planet4589). Medical devices, EVs and robot joints need the same tight-tolerance metal parts, which widens the buyer base further.
- **Headwinds: A new precision plant has to be built and certified before it wins volume work, so capex runs well ahead of revenue.** AS9100, the aerospace quality standard, typically takes 6-12 months, and approval at each buyer takes about nine months to two years.
- **Commercial work brings its own approvals, because aircraft and engine makers also require Nadcap accreditation for special processes.** Nadcap is an industry-run program that audits processes such as heat treating, welding and inspection, along with a first-article approval for each new part.

- **Commercial buyers push hard on price, and a few buyers control most of the volume.** Spirit AeroSystems, Boeing's largest structures supplier, booked $585M of losses on fixed-price contracts in one quarter of 2025 (Reuters), and Boeing bought it back that December (Boeing). Two airframers and a few engine makers place most orders, so a supplier tied to one carries its problems.
- **Defense money arrives slowly and depends on Congress, even in years when budgets rise.** Carpenter Technology reported that its defense bookings fell because customers were "not receiving funds" (Carpenter earnings call).
- **Why we should focus here: An approved supplier keeps the work for programs that can run 50 years, primes pay a premium for fast delivery, and approved incumbents earn strong margins, so the prize at the end of approval is real.** New companies can reach it by building their own plants, by buying certified shops from retiring owners at about seven times annual earnings (claim), by running a franchised network in which franchisees buy the machines, or by hosting cells inside a buyer's plant, and each model trades some margin for less capital.
- **The strongest new precision companies sell to both defense and commercial buyers, because commercial aerospace, power and chip equipment bring larger and steadier volumes that do not depend on appropriations.** That mix keeps machines busy between defense orders and spreads the cost of approvals.

**Example companies:** Hadrian, Isembard, Divergent, Freeform, Amca, Atomic Industries, Fabri, Layup, Hybron, Rangeview, Furientis, Helicon, Westmag, Re:Build, Anduril, Precision Castparts, Howmet

## C. Specialty Chemicals

### Producers (hybrid)

- **Tailwinds: Drug companies are moving discovery chemistry out of China because relying on it has become a supply chain, intellectual property and national-security concern.** We limit this category to automated labs that make new molecules for drug programs. BIOSECURE, a law passed in December 2025, limits the use of certain Chinese biotech suppliers (Baker McKenzie), and 79% of 124 biotechs surveyed had a China-based contract manufacturer (BIO).
- **The government is adding pressure on the largest Chinese chemistry suppliers that US drug companies use.** In June 2026 the Pentagon added WuXi AppTec, the largest Chinese drug research and manufacturing firm, to its list of Chinese military companies (Reuters).
- **AI that designs molecules and robots that make them bring down the cost of doing this work in the US and Europe.** Chemify, a Glasgow-based lab, says it already works with 8 of the 20 largest drug companies (claim, SCI).
- **Headwinds: US chemists cost several times Chinese rates, so these labs have to automate to reach offshore prices.**
- **The category is young, labs have not disclosed revenue, and drug companies are building their own AI labs.** Eli Lilly and NVIDIA plan to invest up to $1B over five years in a joint AI lab for drug discovery (Lilly and NVIDIA, January 2026).
- **The legal push away from China can soften, which would weaken the main reason to switch suppliers.** In August 2026 a federal judge blocked the Pentagon's listing of WuXi AppTec while the case proceeds (Reuters).
- **Why we should focus here: Drug companies want a domestic or allied source for discovery chemistry, and they now have a legal reason to move work out of China.** These labs work at milligram-to-gram scale and do not develop their own drugs. They are paid per compound delivered, or through milestones and royalties if a molecule advances. Each lab costs tens of millions to build, but its reaction data improves its designs with every run. If automation brings unit cost close to offshore prices, the category can take a meaningful share of work that is leaving China.

**Example companies:** Chemify, Warp Sciences

## D. Advanced Materials

### Producers

- **Tailwinds: China makes about 85% of battery cathode material and over 98% of the cheaper LFP type, so battery makers for EVs, grid storage and defense need another source for resilience and security.** The IEA expects China still to hold about three-quarters of cathode production in 2035, with the US and Europe together at about 10% (IEA).
- **Data centers are adding power demand quickly, and cooling takes a growing share of it, which creates demand for new cooling materials.** Data centers are on track to use about 945 TWh by 2030, up from 415 TWh in 2024, and cooling takes 7% to over 30% of a site's power (IEA).
- **AI design aims to shorten how long it takes to bring a new material to market.** A new material has typically taken 10 to 20 years and more than $100M to develop (claim).
- **Headwinds: Customers take one to five years to qualify a new material, so revenue arrives slowly even after the material works.**
- **Owned cathode plants cost hundreds of millions of dollars and compete on Chinese prices.** Umicore, a Belgian leader, saw its battery-materials unit lose money on €436M of revenue (company filings).
- **US demand for some of these materials is softer than forecast as EV sales slow.** The share of LFP batteries in US EVs almost halved, and makers moved capacity to grid storage (IEA).
- **Why we should focus here: These companies design the material with AI in their own lab, then make it and sell it by the kilogram or ton under supply deals, so the company that designs and makes the material keeps the margin, while contract manufacturing keeps the capital need far below that of a refinery.**

**Example companies:** Orbital Materials, Mitra Chem, Chemix, Albemarle, Umicore, Corning

### Materials Discovery Labs

- **Tailwinds: AI and robotic labs can now make and test new materials far faster than human teams.** The leading labs report making more than a thousand alloys in about six months (claim).
- **Non-Chinese magnets and battery materials, stronger alloys for defense and better cooling for data centers all require new materials.** That means the buyers for a successful discovery already exist and are looking for suppliers.
- **Headwinds: These labs do not yet sell material at volume, and their revenue is small next to their valuations.** Revenue today comes from paid R&D, licensing and research awards of around $1M.
- **Finding a material faster does not shorten the years customers take to qualify it.**
- **Investors note that materials science has not yet had its AlphaFold moment, meaning a single AI breakthrough like the model that predicted protein structures (claim).**
- **Why we should focus here, on a longer time horizon: If these labs work, they shorten the slowest part of the materials chain, which is finding a material worth qualifying.** To sell at volume they usually go through a partner that scales the material up and sells it, so a lab earns a fee or royalty unless it builds its own plant.

**Example companies:** Radical AI, CuspAI, Lila Sciences, Periodic Labs, Aionics, Mattiq, Dunia Innovations, Entalpic, Discovered Materials, Kebotix, Citrine Informatics, Uncountable, Albert Invent, Microsoft Discovery

## E. Enabling Software & Robotics

### Enablers

- **Tailwinds: Reshoring only works if US plants can run with fewer skilled people, because about 1.9M US manufacturing jobs are projected to go unfilled by 2033 (Deloitte).** Fully loaded US factory labor costs about $46.30 an hour, while the price of an industrial robot fell from about $46,000 in 2010 to about $23,000 in 2022 (Emergence Capital), so automation pays back faster each year.

- **China installs more industrial robots than the rest of the world combined, and its lead is growing.** China installed 354,226 industrial robots in 2025, 59% of the 603,307 installed worldwide and up 20% from a year earlier, while the US installed 38,428, up 12% (IFR).
- **China is also building its own robot industry, so it is less dependent on Western suppliers each year.** Chinese brands supplied about 30% of robots installed in China in 2019-21 and more than half since 2024 (IFR), and a 2023 government plan for humanoid robots set targets for mass production by 2025 and a secure domestic supply chain by 2027 (MIIT).
- **Washington is starting to shut Chinese robots out of sensitive US sites, which gives US robot and software makers a protected market.** In June 2026 the Pentagon added the Chinese robot maker Unitree to its list of Chinese military companies (Reuters), and on July 28, 2026 the FCC added foreign-made mobile and humanoid robots, including those from allied countries, to its Covered List, which blocks new models from US sale without a conditional approval (Cleary Gottlieb). The House-passed FY27 defense bill would also bar the Pentagon from buying humanoid robots made or controlled by China (H.R. 8800, Sec. 163), though that provision is not yet law.
- **Headwinds: Sales to plants are slow because manufacturers spend cautiously and take months to approve new systems.** Manufacturers expect to raise capital spending by only 2.6% over the next year, and 63% plan to import machinery (NAM).
- **Incumbent vendors such as Siemens and Rockwell Automation bundle software with hardware they have already installed.** Rockwell earns a 49.5% gross margin (Rockwell results), so a startup has to beat a vendor the plant already trusts.
- **Robots are not yet reliable enough for most factory work, since robot-learning success is 59-66% on short tasks against the 99.9% or more that a factory needs, and most humanoids remain pilots (IFR).**
- **The robot arm is only part of the cost, because integrators put the arm at roughly a fifth to under half of an installed cell, with fixtures, safety and programming making up the rest (integrator cost breakdowns).** A Boeing strategy executive told us the net saving after machine costs is 20-30% at best (claim).
- **US startups depend on foreign robot hardware, which limits how much of each sale they keep.** Most robots used in the US are imported from Japan and Europe (IFR), and the German maker KUKA has been owned by China's Midea since 2017 (Reuters).
- **Why we should focus here: China's robot lead means its factories get cheaper and faster every year, so the cost gap with US plants widens unless US plants automate too.** BCG finds automation can cut labor needs by 30-60%, yet a new aerospace and defense plant still costs less to run in China (BCG).
- **Reshoring only pays with automation, because the US lacks the workers and the low wages to copy China's labor-heavy factories.** Much of the returning work is high-mix and low-volume, with defense shops running batches of 10 or 20 parts (NDIA), so the winning systems make a small batch worth automating.
- **The US opportunity is in the software, AI and integration that sit on top of the robot, because the arms themselves come from Japan, Europe and China.** The products that matter program robots from a drawing, schedule plants, inspect parts and run robot cells as a service. Security rules add demand for this layer, because defense plants need software and connected robots that no Chinese company can reach remotely.
- **Buyers pay when the system adds output from machines they already own or covers a shift they cannot staff.** By our estimate, one operator on two shifts costs about $185,000 a year, so a $350,000 cell covering both shifts pays back in about two years. Cutting machines in US plants run only about a third of the time (FTI Consulting), so software that raises their use earns even more.
- **Software and robotics companies do not have to build factories, so they can grow faster than plant owners.** We should back those paid on measured output, by subscription or by the cell-hour, starting in defense and aerospace shops.

**Example companies:** Lumaril, Arda, Mind Robotics, GrayMatter Robotics, CloudNC, AI Robotics, Bright Machines, Rapta, Allus, Faraday, Siemens, Rockwell Automation, Honeywell, AspenTech, Emerson